Experiment to understand the data preprocessing in Data science

In [4]:
```
import numpy as np
import pandas as pd
```

In [5]:
```
df = pd.read_excel("pre_process_datasample - pre_process_datasample.xlsx")
df
```

Out[5]:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

In [6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     str
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     str
dtypes: float64(2), str(2)
memory usage: 452.0 bytes
```

In [7]:
```
df.Country.mode()
```

Out[7]:
```
0    France
Name: Country, dtype: str
```

In [8]:
```
df.Country.mode()[0]
```

Out[8]:
```
'France'
```

In [9]:
```
type(df.Country.mode())
```

Out[9]:
```
pandas.Series
```

In [10]:
```
df["Country"] = df["Country"].fillna(df["Country"].mode()[0])
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Salary"] = df["Salary"].fillna(round(df["Salary"].mean()))
df
```

Out[10]:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

In [11]:
```
pd.get_dummies(df.Country)
```

Out[11]:

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	False	True	False
9	True	False	False

In [12]:
```
updated_dataset = pd.concat(
    [pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
    axis=1
)
updated_dataset
```

Out[12]:

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	False	True	False	50.0	83000.0	No
9	True	False	False	37.0	67000.0	Yes

In [13]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   Country    10 non-null     str    
 1   Age        10 non-null     float64
 2   Salary     10 non-null     float64
 3   Purchased  10 non-null     str    
dtypes: float64(2), str(2)
memory usage: 452.0 bytes
```

In [14]:
```
updated_dataset["Purchased"] = updated_dataset["Purchased"].replace(
    ['No', 'Yes'], [0, 1]
)
```

In [15]:
```
updated_dataset
```

Out[15]:

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	False	True	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

In [ ]: